

Curation Confidence And Hierarchy Spec

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Status: Design proposal (no implementation changes in this doc)

Why this exists

The current curation flow can produce compounding errors:

- Weak or synthetic fields (especially `title`) can influence classification.
- Room assignment is overloaded as the primary style taxonomy.
- A single keyword can force a room assignment even when item context is mixed or ambiguous.

This document defines a more reliable hierarchy and confidence model.

Modeling stance

Hierarchy is useful, but exclusivity should be earned, not assumed.

- Treat most memberships as non-exclusive by default.
- Allow multiple valid memberships at the same level when the asset genuinely supports them.
- Only collapse to a single value when the evidence strongly supports exclusivity.
- Prefer `unknown` / `ambiguous` over false certainty.

This matters because many inspiration assets are hybrid:

- an outdoor dining scene is both `outdoor_zone` and `function=dining`
- a close-up of a sink may belong to bathroom intent while also being a product-focused item
- a ZIP System reference can be both an envelope concern and a specific system/product focus

Primary goals

1. High confidence discrimination between:
 - `style_product_decor`
 - `construction_concern`
2. High confidence categorization within each track:
 - Construction: concern themes and planning structure
 - Style: room context plus additional peer-level dimensions

Evidence from current corpus

From scope `trriage_status` in `(pending, keeper)`:

- Candidates: 4663

- If `title` is blanked in heuristic mode:
 - 69 items change track
 - 99 items change room
 - 46 / 3050 style items change room
- 17 items are assigned `dining_room` while also carrying outdoor signals (`patio`, `garden`, `outdoor`, etc.).
- Missing context:
 - 150 missing board
 - 64 missing labels
 - 47 missing both board and labels

Reference: `data/exports/provenance_audit_20260305.json`

Core diagnosis

Current style grouping is effectively single-axis (`style -> room`).
Real data is multi-axis and often non-exclusive.

A patio dining scene is not an interior dining room, even if the function is dining.
The missing distinction is between:

- Space context (indoor/outdoor/non-spatial)
- Function (dining/cooking/sleeping)
- Subject type (full room/product/detail)
- Product or system focus (named product families, equipment classes, assemblies)

Proposed hierarchy model

1) Top gate (hard)

- track:
 - `style_product_decor`
 - `construction_concern`
 - `irrelevant`

2) Style track axes (peer-level)

Do not force all style items into `room`.

Axis	Purpose	Example values
<code>space_context</code>	Physical context of scene	<code>interior_room</code> , <code>outdoor_zone</code> , <code>transition_space</code> , <code>whole_home</code> , <code>non_spatial</code>

Axis	Purpose	Example values
<code>subject_type</code>	What the asset primarily is	<code>full_space_scene</code> , <code>vignette_styling</code> , <code>single_product</code> , <code>material_finish</code> , <code>architectural_detail</code> , <code>plan_drawing</code>
<code>function</code>	Functional intent	<code>dining</code> , <code>cooking</code> , <code>sleeping</code> , <code>bathing</code> , <code>storage</code> , <code>circulation</code> , <code>utility</code> , <code>entertaining</code>
<code>room</code> (conditional)	Interior room only when valid	<code>kitchen</code> , <code>bathroom</code> , <code>bedroom</code> , <code>living_room</code> , <code>dining_room</code> , ...
<code>product_focus</code>	Specific product or fixture class when the item centers on one	<code>range</code> , <code>refrigerator</code> , <code>dishwasher</code> , <code>sink</code> , <code>toilet</code> , <code>tub</code> , <code>faucet</code> , <code>lighting_fixture</code> , <code>vanity</code> , <code>hardware</code>
<code>design_facets</code>	Reusable style descriptors	<code>styles</code> , <code>palette</code> , <code>materials</code> , <code>element focus</code> , <code>decor motifs</code>

Rule: `room` is assigned only if `space_context=interior_room` and `subject_type=full_space_scene` (or strong interior scene evidence).

`product_focus` is not subordinate to `room`. A sink can be bathroom-related without requiring a full bathroom room assignment.

3) Construction track axes (peer-level)

Axis	Purpose	Example values
<code>concern_domain</code>	Existing concern theme	<code>site_exterior</code> , <code>envelope</code> , <code>structure</code> , <code>mep</code> , <code>plans_code_permits</code> , <code>interiors_execution</code>
<code>project_phase</code>	Where concern applies	<code>concept</code> , <code>design</code> , <code>permit_code</code> , <code>procurement</code> , <code>build</code> , <code>commissioning_closeout</code>

Axis	Purpose	Example values
<code>trade_system</code>	Trade/system ownership	<code>structural</code> , <code>envelope</code> , <code>mechanical</code> , <code>electrical</code> , <code>plumbing</code> , <code>site</code> , <code>millwork_finish</code> , <code>multi_trade</code>
<code>concern_class</code>	Concern type semantics	<code>risk</code> , <code>decision</code> , <code>requirement</code> , <code>checklist</code> , <code>reference_example</code>
<code>product_system_focus</code>	Specific product family, named assembly, or equipment class under discussion	<code>water_heater</code> , <code>tankless_water_heater</code> , <code>zip_system</code> , <code>window_system</code> , <code>waterproofing_membrane</code> , <code>insulation_system</code> , <code>roofing_system</code> , <code>duct_system</code>

`trade_system` stays broad. `product_system_focus` captures the specific thing being evaluated or repeated across references.

Confidence model (provenance-aware)

Principle

Not all fields are equal. Each field needs both:

- `origin` (where it came from)
- `trust_weight` (how much influence it should have)

Suggested signal tiers

Tier 1 (highest trust):

- User board intent when board semantics are clear
- Human-reviewed labels/overrides

Tier 2:

- Source-native structured metadata (when stable)
- Strong AI vision tags with high confidence and low contradiction

Tier 3:

- Imported text titles/descriptions
- AI-generated summaries
- Heuristic keyword expansions

Decision logic

Use weighted voting per axis, not first-hit keywords.

```
score(axis_value) = sum(signal_weight * signal_confidence *  
provenance_weight)
```

Then:

- Accept if top score exceeds threshold and separation margin.
- Else assign **unknown** / **ambiguous**.

No forced assignment when evidence conflicts.

Important: this scoring is per axis, not one global winner-take-all label. An asset may carry multiple accepted memberships across peer-level axes.

Proposed classification flow

flowchart TD

```
A["Collect Candidate Signals"] --> B["Track Gate: style vs construction vs irrelevant"]  
B -->|style| C["Style Axis Inference"]  
B -->|construction| D["Construction Axis Inference"]  
C --> C1["Infer space_context"]  
C1 --> C2["Infer subject_type"]  
C2 --> C3["Infer function"]  
C3 --> C4["Infer room only when interior scene is valid"]  
C4 --> C5["Infer product_focus when item centers on a fixture or appliance"]  
C5 --> C6["Attach design_facets"]  
D --> D1["Infer concern_domain"]  
D1 --> D2["Infer project_phase"]  
D2 --> D3["Infer trade_system"]  
D3 --> D4["Infer product_system_focus"]  
D4 --> D5["Infer concern_class"]  
C6 --> E["Confidence + Conflict Resolution"]  
D5 --> E  
E --> F["Export with unknown/ambiguous allowed"]
```

Conflict policy

When signals disagree:

- If outdoor evidence is strong and dining evidence is strong:
 - `space_context=outdoor_zone`
 - `function=dining`
 - `room=null`
- If an item is clearly a product close-up:
 - allow `product_focus=<value>`
 - do not force `room`

- If a construction item clearly references a named system or equipment family:
 - allow `product_system_focus=<value>`
 - do not force that field to substitute for `concern_domain` or `trade_system`
- If no dominant winner:
 - assign `unknown` / `ambiguous`
 - preserve evidence vector for human review

Do not coerce to nearest room.

Example outcomes

Example A: Patio table scene

Current misfire: `room=dining_room`

Proposed:

- `track=style_product_decor`
- `space_context=outdoor_zone`
- `subject_type=full_space_scene`
- `function=dining`
- `room=null`
- `design_facets=[...]`

Example B: Bathroom faucet close-up

Proposed:

- `track=style_product_decor`
- `space_context=interior_room` (weak)
- `subject_type=single_product`
- `function=bathing`
- `product_focus=sink or faucet`
- `room=null` (not enough full-room evidence)

Example C: Foundation detail and waterproofing note

Proposed:

- `track=construction_concern`
- `concern_domain=structure or envelope` (by evidence)
- `project_phase=build`
- `trade_system=structural or envelope`
- `product_system_focus=waterproofing_membrane` or other named assembly if present
- `concern_class=risk or requirement`

Example D: ZIP System reference

Proposed:

- track=construction_concern
- concern_domain=envelope
- trade_system=envelope
- product_system_focus=zip_system
- concern_class=reference_example or requirement

Example E: Range appliance inspiration

Proposed:

- track=style_product_decor
- subject_type=single_product
- function=cooking
- product_focus=range
- room=null unless the asset is also a valid full kitchen scene

Data contract additions (recommended)

Per asset classification payload should include:

- track
- track_confidence
- space_context, space_context_confidence (style only)
- subject_type, subject_type_confidence (style only)
- function, function_confidence (style only)
- room (nullable), room_confidence (style only)
- product_focus[], product_focus_confidence (style only; multi-valued allowed)
- concern_domain, project_phase, trade_system, concern_class (+ confidences, construction only)
- product_system_focus[], product_system_focus_confidence (construction only; multi-valued allowed)
- is_ambiguous (boolean)
- evidence object with weighted contributors

Per source field provenance metadata should include:

- title_origin (imported, ai_suggested, title_audit, manual_edit, etc.)
- title_last_mutated_at
- title_mutator
- Similar metadata for description, board, and key labels where possible

Success metrics

Track discrimination

- Precision/recall for style vs construction on reviewed sample
- Target: materially lower cross-track contamination than current baseline

Style categorization quality

- Outdoor dining incorrectly forced into interior room should trend toward zero.
- Reduced room assignment when `subject_type != full_space_scene`.
- Reviewer agreement rate on `space_context` and `subject_type`.
- Reviewer agreement rate on `product_focus` for fixture/appliance-led style items.

Construction categorization quality

- Reviewer agreement on `concern_domain`
- Higher planning usefulness from `project_phase` and `concern_class`
- Reviewer agreement on `product_system_focus` for named assemblies and equipment classes.

Rollout strategy

1. Phase 1: Provenance instrumentation
 - Add field-origin metadata and mutation history pointers.
2. Phase 2: Multi-axis inference without changing export grouping defaults
 - Run side-by-side with existing room-only output.
3. Phase 3: Confidence gating and ambiguity handling
 - Enable `unknown` paths.
4. Phase 4: Export and UI updates
 - Present room only when valid; preserve function/context axes.

Immediate recommendation

Before additional ranking tuning:

1. Lock taxonomy design for the multi-axis model.
2. Implement provenance-aware confidence scoring.
3. Introduce ambiguity handling and stop forced room assignment.
4. Use pairwise only after trusted categorization, not as a fix for categorization errors.